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COURSE : CSA0927 – PROGRAMMING IN JAVA

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner scanner = new Scanner(System.in);
        try{
            System.out.println("Enter the first integer(numerator):");
            int numerator = scanner.nextInt();
            System.out.println("Enter the second integer(denominator):
            int denominator = scanner.nextInt();
            int result = numerator / denominator;
            System.out.print("Result:"+result);
        }
        catch(ArithmeticException e){
            System.out.println("Error:Cannot divide by zero");
        }
        catch(Exception e){
            System.out.println("Error: An unexpected error occured.");
        }
        finally{
            scanner.close();
        }
    }
}
```

Input

```
10
0
```

Output

```
Enter the first integer(numerator):
Enter the second
integer(denominator):
Error:Cannot divide by zero
```

Q2 **Handling ArrayIndexOutOfBoundsException** 10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = {10, 20, 30, 40, 50};
        System.out.println("Enter an index: ");
        try {
            int index = sc.nextInt();
            System.out.println("Element at index "+index+ "is:" +arr[i
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Error: Index is outside the valid rang
        } finally {
            sc.close();
        }
    }
}
```

Input

```
5
```

Output

```
Enter an index:
Error: Index is outside the valid
range (0 to 4).
```

Q3 **Handling NumberFormatException** 10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
import java.util.Scanner;
public class NumberFormatExceptionExample{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a number as a string:");
        String input = sc.nextLine();
        try {
            int num = Integer.parseInt(input);
            System.out.println("Converted Integer value: "+num);
        } catch (NumberFormatException e){
            System.out.println("Invalid input! Please enter a valid in
        }
        sc.close();
    }
}
```

Input

```
12.6
```

Output

```
Enter a number as a string:
Invalid input! Please enter a valid
integer.
```

Q4 **Handling Multiple Exceptions** 10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both ArithmeticException and ArrayIndexOutOfBoundsException using multiple catch blocks.

Your Code

```java
import java.util.Scanner;
public class MultipleExceptionHandling{
    public static void main(String[] args){
        Scanner scanner = new Scanner(System.in);
        int[] results = new int[1];
        try {
            System.out.print("Enter first integer: ");
            int num1 = scanner.nextInt();
            System.out.print("Enter second integer: ");
            int num2 = scanner.nextInt();
            int divisionResult = num1 / num2;
            results[0] = divisionResult;
            System.out.println("Result stored in array: " + results[0]
            System.out.println("Attempting to access index 1...");
            System.out.println(results[1]);
        }
        catch (ArithmeticException e){
            System.out.println("Error: Cannot divide by zero! (Arithme
        }
        catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Error: Array index is out of bounds! (
        }
        catch (Exception e) {
            System.out.println("An unexpected error occurred: " + e.ge
        }
        finally {
            scanner.close();
            System.out.println("Program execution completed.");
        }
    }
}
```

Input

```
10
0
```

Output

```
Enter first integer: Enter second
integer: Error: Cannot divide by
zero! (ArithmeticException)
Program execution completed.
```